

Research Statement

Sean Patrick MacBride

My research focuses on exploring the cosmic frontier using electromagnetic and gravitational wave data. This focus is underscored by three distinct pillars: building cutting-edge astronomical instrumentation, leading observing campaigns to search for the electromagnetic counterparts to gravitational-wave mergers, and performing inference on electromagnetic and gravitational-wave data to achieve robust cosmological constraints. These three pillars reflect the approach I aim to bring to **Add Institution**, where each research activity is not viewed as separate, but as three legs of one research program, strongest when pursued together.

TODO

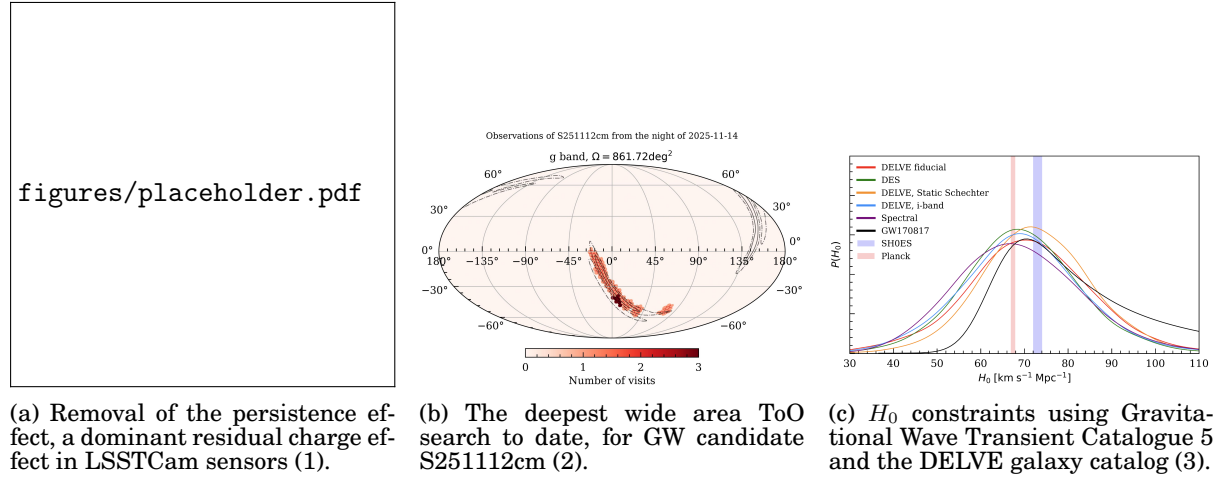


Figure 1: A summary of early career contributions, spanning instrumentation, multi-messenger astrophysics, and standard siren cosmology.

Early Career Achievements: My past work has been in direct support of the three pillars of my research program, and includes commissioning the focal plane array of LSST Camera (LSSTCam), producing the first science result for the Prime Focus Spectrograph (PFS), conducting the first Target-of-Opportunity (ToO) searches for electromagnetic counterparts to gravitational waves using Vera C. Rubin Observatory (Rubin), and measuring H_0 using the Dark Energy Local Volume Exploration (DELVE) galaxy catalog.

Astronomical Instrumentation Development: My instrumentation work has encompassed both photometric and spectroscopic instruments, with a specific focus on focal plane technology. This began with Large Lenslet Array Magellan Spectrograph (LLAMAS), an integral field spectrograph for the Magellan Baade telescope. I validated the optical performance of the diffraction gratings and collimators for the LLAMAS spectrographs, assembled and verified optical tolerance for the 24 spectrograph cameras, and integrated all 2,400 AR-coated fibers into the focal plane with zero fiber loss, before it was commissioned in 2024 (4; 5). After integrating and testing LLAMAS, I shifted my focus from integral-field spectrographs to multi-object spectrographs. While at the University of Michigan-Ann Arbor, I characterized and resolved a common failure mode on Dark Energy Spectroscopic Instrument (DESI) robotic positioners, known as the linear phi failure mode. While testing the DESI positioners, I found that reorienting the robotic positioner to an inverted physical configuration could remove the linear phi behavior, indicating that gravity-sag has a significant influence on the failure mode (6). This expertise with spectroscopic instruments translated effectively to the transcendent photometric imager of the next decade, the LSSTCam. Since 2024, I have served as an LSSTCam Summit Support Scientist, leading

LSSTCam focal-plane commissioning on Cerro Pachón as a key component of the in-kind contribution from Switzerland to the Rubin Project. I tested the CCDs in both the summit clean-room and on the Simonyi Survey Telescope, optimizing the sensor performance, and played a leading role in achieving LSSTCam first photon in April 2024 (see Figure 1a). The work to commission LSSTCam, among other CCD optimizations, is detailed in the invited proceedings of SPIE Astronomical Telescopes + Instrumentation 2026, the most impactful astronomical instrumentation conference, held biennially (1).

Target-of-Opportunity Searches: I have led numerous observing campaigns using different telescopes and facilities, with the goal of detecting to electromagnetic counterparts to gravitational wave merger events. Starting in 2023, I co-led the Dark Energy Survey Gravitational Wave follow-up program (DESGW) program, conducting Dark Energy Camera (DECam) observations, processing raw image data from 13 ToO campaigns to calibrated products, and analyzing processed data in response to International Gravitational Wave Network (IGWN) Observing Run 4 (O4) triggers. When a well-localized gravitational-wave merger candidate was accessible to both Northern and Southern hemisphere instruments, I led a collaboration between DESGW and the Japanese collaboration for gravitational-wave electromagnetic follow-up (J-GEM). In this collaboration, I obtained multi-epoch photometry using DECam, processed the obtained data, and cross-matched this data with spectroscopic observations using PFS at Subaru Telescope, enabling the first public scientific result using PFS (7). Since 2025, I have been leading the Rubin ToO observing program as the Rubin ToO observer. In this role, I have led observing responses to one interstellar object (8), one high-energy neutrino event (later reclassified as terrestrial), and two gravitational-wave merger candidates, including the deepest wide-area gravitational wave ToO search to date ($m_{r,5\sigma} \sim 24.4$ across $\sim 800 \text{ deg}^2$ on four nights, see Figure 1b) (9; 2).

Standard Siren Constraints on the Hubble Constant: While recent observing campaigns have not yielded an electromagnetic counterpart to a gravitational wave merger, I have retained a focus on constraining cosmology using electromagnetic and gravitational wave data, employing the *dark siren* method to measure the expansion rate of the local universe, H_0 . This work began by supporting the preparation of Dark Energy Survey (DES) data, resulting in the first dark siren constraint using a deep photometric catalog to jointly infer gravitational wave population parameters and cosmological parameters (10). DES was accepted as the fiducial result for the most recent IGWN analysis, and demonstrated a 22% improvement in constraining H_0 compared to previous analyses (11). Building on the DES analysis, I am now leading a dark-siren analysis using the DELVE galaxy catalog, where initial results show DELVE is competitive with DES, obtaining a 12% constraint on H_0 when combined with the bright siren GW170817 (see Figure 1c) (3).

Future Research Activities: At **Add Institution**, I will build directly on my current **TODO** work in instrumentation, multi-messenger astrophysics, and standard siren cosmology to develop instruments for next-generation astronomical surveys, maximize the data from Rubin Observatory for bright siren measurements, transform Legacy Survey of Space and Time (LSST) into the standard dark siren galaxy catalog, and lead new observational programs for standard siren cosmology.

Instrumentation Development for the Cosmic Frontier: I will continue in a support-scientist role for LSSTCam as the camera transitions from early operations into sustained LSST survey operations, maintaining the operational knowledge needed to respond quickly to sensor anomalies and focal-plane issues as they arise. Building on my commissioning-era characterization of focal-plane defects, I will deploy a defect-monitoring tool, currently in-development, to `cp_verify`, the calibration-product verification framework used by the LSST Science Pipelines. Bringing this tool from isolated development to a published resource will move defect monitoring from a manual to automated process, directly improving the quality of LSST data.

The central bottleneck for transient astrophysics and standard siren cosmology in the LSST era is not imaging depth, but spectroscopic capacity: both dark and bright siren measurements ultimately require secure redshifts for host galaxies and rapid characterization of transients from LSST's alert stream. To aid this effort, I will expand on my expertise from testing the robotic positioners of DESI to test advanced focal plane technology for future multi-object spectrographs. As part of this focus, I will explore instrument design for a multi-object spectrograph optimized for standard siren cosmology, specialized for the precision and event rates of forthcoming gravitational wave detectors. This instrument would serve two goals; 1) characterize

astrophysical transients associated with potential bright standard sirens, and 2) obtain spectroscopic redshifts from host galaxies to bolster dark siren measurements. Through this dual purpose, the proposed instrument would become a dark energy science machine primed to capitalize on the deluge of data from LSST and future GW observatories. These pathfinding activities would help to inform the proposed Stage-5 Spectroscopic Experiment (Spec-S5), a future, large-scale cosmic frontier experiment. Development and testing of robotic positioners for the focal plane will help shape instrument capabilities so it serves as an effective spectroscopic complement to Rubin, starting in the mid-2030's.

Bright Siren Searches Enabled by Rubin Observatory: I will lead the observing planning, execution, and analysis of the next bright standard siren accessible to Rubin, along with other ToO's pursued during the LSST. While it is difficult to predict the timing of the next bright siren, I will continue to develop and deploy improvements to the ToO system to improve response latency, observing strategy, and system automation to maximize the forthcoming opportunity to detect and characterize these optical counterparts to obtain a well-constrained measure of H_0 . In parallel to operating the Rubin ToO system, I will support the analysis of DESGW candidates identified during IGWN O4, and lead the analysis of DESGW candidates identified during IGWN intermediate runs. This work positions me to sustain and support Rubin's ToO program with additional imaging in other filters or to extended time-frames using a similarly capable photometric imager. This increased temporal sampling and color characterization will play an important role, especially as other priorities of the LSST are balanced against the ToO program. As LSST's ToO program matures, I will work to enhance counterpart characterization by securing spectroscopic follow-up time on other instruments, moving beyond photometric classification to confirm spectral features that can inform the scope of Rubin ToO's and constrain the physical properties of the underlying transients and their host environments. This includes the an ongoing program with J-GEM using PFS, a new program using LLAMAS to characterize promising candidates from the LSST alert stream, and a future program using data from the BOOMBOX spectrograph. **Double check if applying to a Carnegie institution, otherwise swap with SOAR.**

LSST data for Dark Siren Cosmology: I will pursue a cutting-edge program that produces a gold-standard galaxy catalog for dark siren cosmology using LSST data. This

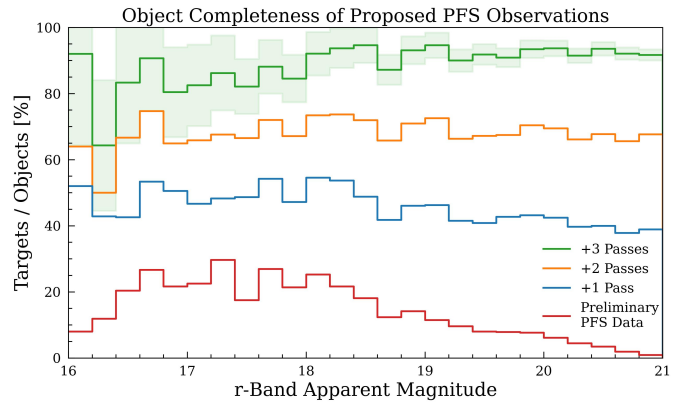


Figure 2: In red, the completeness of PFS objects acquired as part of our ToO campaign (7), with other colors showing the forecasted completeness of proposed observations increasing with each successive pass over the target area. The high completeness obtained over the 75% localization region in 3 passes (one night) demonstrates the utility of current and future multi-object spectrographs as characterization machines for standard siren cosmology.

program proceeds in four stages, each building towards a powerful, world leading measurement. It begins with an ongoing project, focused on characterizing the LSST Data Preview 2 (DP2) galaxy catalog, focusing on robust object classification that is scalable to future LSST data releases. **Could reference Andres paper if it is published soon-ish** Following characterization of DP2, I will finish a forecasting study that quantifies the H_0 precision achievable by combining simulated first-year LSST data with an O4-like gravitational-wave catalog, extending the dark-siren methodology from my recent DELVE measurement to project the constraining power in the LSST era. Superseding DP2 characterization and forecasted precision, I will characterize the first LSST Data Release (DR1) galaxy catalog and produce a dark siren measurement of H_0 . This result will extend the methodology I developed with DELVE to a dataset with substantially greater depth, area, and photometric uniformity, all features that will make it the standard galaxy catalog for dark siren analyses. Building on the DR1 measurement, I aim to combine LSST large-scale structure measurements in a joint cosmological analysis, using cosmic shear and standard sirens to obtain a tighter constraint of H_0 , and extend the constraints to Ω_m .

TODO

New Observational Programs and Analyses for Standard Sirens: As the wealth of gravitational wave data increases alongside a wide variety of new galaxy catalog data, I will explore the frontier of the standard siren method using specialized surveys and forthcoming data releases from other surveys, all while engaging with the Horizon Europe-funded GWCats collaboration, a tight collaboration of researchers focused on using galaxy catalogs and gravitational wave data to constrain cosmology. To observationally assess the frontier of the dark siren method, I will lead a dedicated spectroscopic dark-siren pilot survey, obtaining targeted redshifts within GW localization volumes to assess the limits and enhancements to H_0 constraints using well-localized dark sirens. This survey is led through a continuing collaboration with J-GEM on the PFS instrument, obtaining additional observations for the GW merger pursued in our collaborative DESGW ToO with an expected observation date starting in April 2027 (see Figure 2). In parallel, I will lead the image processing for a similar survey, the Gravitational Wave Photometric Redshift Improvement with Medium-band Observations (GW-Primo) project, to test how medium band observations that improve photometric redshifts can affect H_0 precision. By comparing the results of these two survey programs, I will be able to assess which route, enhanced photometric or spectroscopic, will provide the best constraints on H_0 given limited telescope time and the current sample of well-localized GWs. Similar to DES the The Ultraviolet Near-infrared Optical Northern Survey (UNIONS) survey provides a large area galaxy catalog in the Northern hemisphere, where DES and LSST coverage is not possible. I will prepare the UNIONS galaxy catalog for dark siren cosmology through ongoing work with the UNIONS collaboration, applying the lessons from the DELVE galaxy catalog preparation. I will construct both an isolated UNIONS catalog and a combined galaxy catalog of UNIONS and LSST data for standard siren cosmology, while taking care to control any systematics that come from joining different galaxy catalogs. This combined catalog will span most of the accessible extragalactic sky, including several well localized GWs in the northern hemisphere, becoming a value-added catalog compared to LSST data alone.

These plans form one continuous arc, not separate agendas. The near-term focus on builds the operational standing and analysis pipelines within Rubin, LSST, and LIGO-Virgo-Kagra (LVK), and provides the basis for executing on the longer-term program. That foundation will enable impactful standard-siren measurements, a pioneering specialized survey for gravitational wave cosmology, combined-probe

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analyses in LSST-DESC, and instrumentation development to maximize the 2030's using both spectroscopic and photometric instruments.

References: A full publication list is available on Google Scholar and NASA ADS (h-index 8 as of August 2026).

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